5 **(a)** Define electric field strength.

.....

.....

.....[1]

(b) Fig. 5.1 shows two plates A and B, a distance 30 mm apart in vacuum, with A at a potential of –4.5 V and B at zero potential.

Electrons are emitted from B with upwards velocity, *u*.

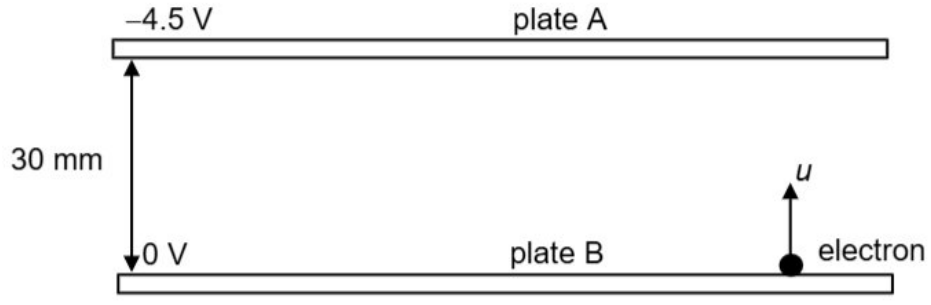


Fig. 5.1

(i) Calculate the electric field strength between the plates.

electric field strength = ... V m^{-1} [2]

(ii) Determine the minimum value of *u* in order for electrons to reach plate A.

minimum value of *u* = ... m s^{-1} [2]

(iii) State and explain how, if any, the value in **(b)(ii)** will change if the distance between the plates is doubled while keeping the plate potentials unchanged.

.....

.....

.....

.....[2]

(c) An uncharged capacitor of capacitance 160 μF is charged to a potential difference of 200 V.

(i) Determine the charge on the capacitor.

charge = ... C [1]

(ii) This charged capacitor is now connected across a resistor until the potential difference decreases to 100 V. Calculate the energy dissipated in the resistor.

energy dissipated = ... J [2]

[Total: 10]